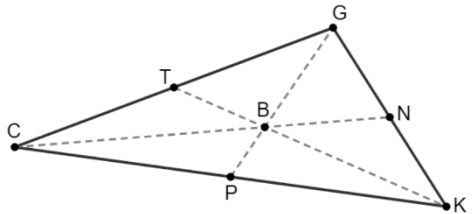


## 11.7 Centroid of a Triangle

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                          |                                                                                      |                                                                                                                                              |                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark Focus     | MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.<br><br>MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment. |                                                                                      |  Learning Targets                                         | <ul style="list-style-type: none"><li>I can find the centroid of a triangle.</li><li>I can use the relationship of the distance a centroid is from a vertex to the midpoint of the opposite side to solve problems.</li></ul> |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                              |                                                                                      |                                                                                                                                              | Working Toward                                                                                                                                                                                                                |
|                                                                                                       | MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.                                                                                                                                                                                   |                                                                                      |                                                                                                                                              | MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.                                                                        |
|  Essential Questions | <ul style="list-style-type: none"><li>How do you find the centroid of a triangle?</li><li>What is the relationship between the distance from the vertex to the centroid, and the distance from the centroid to the midpoint?</li></ul>                                                                   |                                                                                      |                                                                                                                                              |                                                                                                                                                                                                                               |
|  Lesson Narrative   | In this lesson, students explore how to construct medians and centroids of triangles. Then, they discover that the length from the vertex to the centroid is twice the length from the centroid to the midpoint. Finally, they apply their knowledge of this relationship to solve problems.             |                                                                                      |                                                                                                                                              |                                                                                                                                                                                                                               |
|  Component Summary | 11.7.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                 | Order angles from least to greatest based on given side lengths ( <i>SE p. 214</i> ) | 11.7.3 Guided Instruction (10 min.)                                                                                                          | Discover length from vertex to centroid is twice length from centroid to midpoint ( <i>SE p. 217</i> )                                                                                                                        |
|                                                                                                       | 11.7.2 Exploration (20 min.)                                                                                                                                                                                                                                                                             | Explore how to construct medians and centroids of triangles ( <i>SE p. 214</i> )     | 11.7.4 Your Turn! (5-10 min.)                                                                                                                | Practice solving problems involving medians and centroids; analyze error in sample work ( <i>SE p. 218</i> )                                                                                                                  |
|                                                                                                       | 11.7.5 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                                                 | Summarize lesson for hypothetical student who missed class ( <i>SE p. 219</i> )      |                                                                                                                                              |                                                                                                                                                                                                                               |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                 |                                                                                      | Materials                                                                                                                                    | Additional Preparation                                                                                                                                                                                                        |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Median of a Triangle</li><li>Centroid of a Triangle</li></ul> <u>Key Terms:</u> <ul style="list-style-type: none"><li>Vertex</li><li>Midpoint</li></ul>                                                                                          |                                                                                      | <ul style="list-style-type: none"><li>Patty Paper</li><li>Compass</li><li>Triangle Manipulative</li><li>(Optional) Colored Pencils</li></ul> | <ul style="list-style-type: none"><li>Prior to the lesson, consider cutting out triangle manipulatives, made of cardboard or posterboard, to provide a hands-on approach to exploring the location of the centroid.</li></ul> |

|                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher<br>Tips              | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>Students may struggle with solving problems where the distance from the vertex to the centroid is twice the distance from the centroid to the midpoint.</li><li>Students may struggle with constructing perpendicular bisectors.</li></ul>                                                                                                                                                                      | <b>Things to Consider</b> <ul style="list-style-type: none"><li>Consider creating and displaying an anchor chart with the definition of median and centroid of a triangle or adding the terms to a classroom Word Wall.</li><li>Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the location of the centroid.</li><li>If time is an issue, consider assigning 11.7.4 Your Turn! for homework.</li></ul> |
|                                                                                                                   | <b>Literacy and Language Routines</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Mathematical Thinking and Reasoning (MTR) Standard</b>                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                                                                   | <b>Compare and Connect</b> <p>During 11.7.2 Exploration, students have an opportunity to compare the distance from the vertex to the centroid to the distance from the centroid to the midpoint. After constructing the centroid, prompt students to mark each distance using patty paper. Ask students what relationship they notice and prompt them to test their conjectures with each median in the triangle.</p>                                                              | <b>MA.K12.MTR.4.1: Discussion and Reflection</b> <p>In 11.7.4 Your Turn!, students have the opportunity to demonstrate their mastery of the learning target by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors themselves in the future.</p>                                                          |
| <br>Differentiate<br>Instruction | <p>Consider allowing students to find the balancing point in 11.7.2 Exploration by providing students with a triangular manipulative made from a piece of cardboard or poster board. Students can balance the triangular manipulative on their pencil to discover the location of the centroid. This provides students with visual and kinesthetic entry points. Encourage students to discuss their discoveries with a partner to provide entry points for auditory learners.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Lesson Notes:                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                            |



## 11.7.1 Warm-Up: Bell Work

**Component Overview:** Students use the given triangle side lengths to order the angle measurements from least to greatest. This activity is based on the work that students completed in Lesson 6. Provide support to any students who struggle.



### 11.7.1 Warm-Up: Bell Work

- Given  $\triangle BMT$ , where  $MB = 7$ ,  $MT = 12$ , and  $BT = 9$ , list the measures of the angles in order from least to greatest.  
 $m\angle MTB, m\angle BMT, m\angle MBT$



Challenge students who finish early to create their own similar problems, swap problems with a peer, and solve each other's problems. Remind students to use the Triangle Inequality Theorem to make sure their side lengths in their problems can be used to construct a triangle.

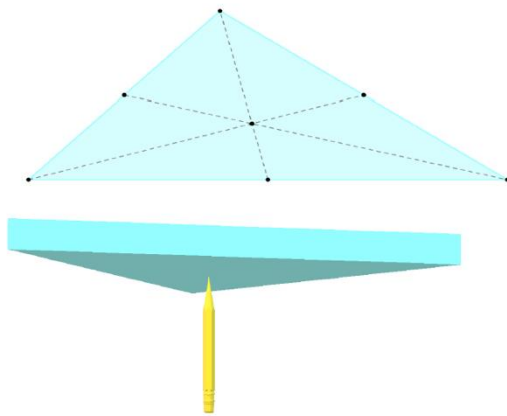
## 11.7.2 Exploration: Centroid of a Triangle

**Component Overview:** Students explore how to construct the median and the centroid of a triangle. Students have previously worked with medians and constructed perpendicular bisectors. However, this will be the first time students have been exposed to the centroid.



### 11.7.2 Exploration: Centroid of a Triangle

1. All polygons have a balancing point. The diagrams show the balancing point of a triangle.



- a. What do you notice?

*Answers will vary. I notice that the balancing point is inside the triangle.*

- b. Discuss with your partner how you think the balance point was found. Summarize your discussion.

*Answers will vary. In the bottom picture it is where the triangle will balance on the pencil. In the top picture, it is the point of concurrency of the lines drawn from each vertex to the midpoint of the opposite side.*

A **median of a triangle** is a segment that joins any vertex of the triangle to the midpoint of the opposite side.



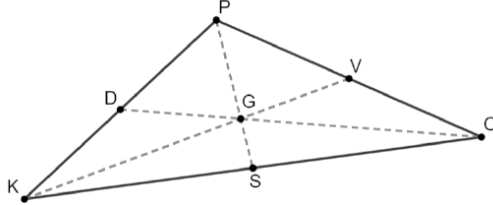
*Provide kinesthetic and visual entry points by giving students triangular manipulatives (made from cardstock/cardboard or triangular snack chips, such as Doritos). Prompt students to find the balancing point with their pencils. Connect to students' prior experiences balancing on a see saw.*



- *How did you find the balancing point?*
- *Where is the balancing point located in relation to the triangle's vertices and sides?*

## 11.7.2 Exploration: Centroid of a Triangle (continued)

2.  $\triangle KPC$  is shown, where point  $G$  is a point of concurrency.



Discuss with your partner how to find the midpoints of the triangle's sides using each method. Complete the table.

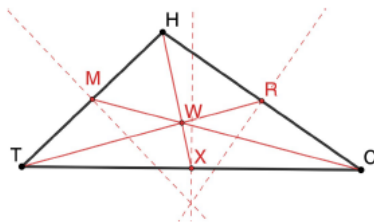
| Patty Paper                                                                                                            | Compass                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Answers will vary. Fold each side so that two vertices line up. Crease the side at the midpoint. Repeat for each side. | Answers will vary. Use the compass to find each side's perpendicular bisector and the midpoint will be at the point of intersection. |



- What are the similarities and differences between how you find the midpoints of triangles using patty paper versus using a compass?
- Which method do you prefer and why?

## 11.7.2 Exploration: Centroid of a Triangle (continued)

3.  $\triangle HCT$  is shown.



- a. Use the steps to construct the centroid of  $\triangle HCT$ .  
*Construction shown above.*

|                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------|
| Step 1: Trace $\triangle HCT$ on patty paper.                                                                                       |
| Step 2: Construct the perpendicular bisector of $\overline{HT}$ to find its midpoint, $M$ .                                         |
| Step 3: Draw the median from $M$ to the opposite vertex, $\angle HCT$ .                                                             |
| Step 4: Construct the perpendicular bisector of $\overline{HC}$ to find its midpoint, $R$ .                                         |
| Step 5: Draw the median from $R$ to the opposite vertex, $\angle HTC$ .                                                             |
| Step 6: Construct the perpendicular bisector of $\overline{TC}$ to find its midpoint, $X$ .                                         |
| Step 7: Draw the median from midpoint $X$ to the opposite vertex, $\angle THC$ .                                                    |
| Step 8: Draw point $W$ , the point of concurrency, where medians $\overline{RT}$ , $\overline{XH}$ , and $\overline{MC}$ intersect. |

- b. What do you notice about the location of point  $W$  in  $\triangle HCT$ ?

*Answers will vary. Point  $W$  is on the inside of the triangle and located at the intersection of all the medians of triangle  $HCT$ .*



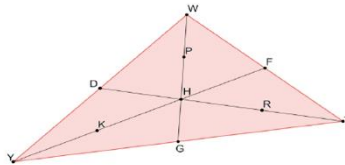
Consider creating an anchor chart with the steps required to construct a centroid. Add a third column with a visual diagram of each step of the construction. Color code and label the diagrams to provide students with a visual entry point.



Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the location of the centroid.

## 11.7.2 Exploration: Centroid of a Triangle (continued)

4.  $\triangle YWT$  is shown, with medians  $\overline{WG}$ ,  $\overline{TD}$ , and  $\overline{YF}$ . Point  $P$  is the midpoint of  $\overline{WH}$ , point  $R$  is the midpoint of  $\overline{TH}$ , and Point  $K$  is the midpoint of  $\overline{YH}$ .



- a. Copy the first segment using patty paper. Use the copy to determine the relationship the first segment and the point have with the second segment. Complete the table.

| First Segment   | Point | Second Segment  | Relationship                                           |
|-----------------|-------|-----------------|--------------------------------------------------------|
| $\overline{DH}$ | $R$   | $\overline{TH}$ | $\overline{TH}$ is twice the length of $\overline{DH}$ |
| $\overline{FH}$ | $K$   | $\overline{YH}$ | $\overline{YH}$ is twice the length of $\overline{FH}$ |
| $\overline{GH}$ | $P$   | $\overline{WH}$ | $\overline{WH}$ is twice the length of $\overline{GH}$ |

The **centroid of a triangle** is the intersection of the three medians of the triangle (each median connecting a vertex with the midpoint of the opposite side).

- b. Write a conjecture about the distance a centroid is from the vertex and the distance a centroid is from the midpoint of the opposite side.  
*The centroid is double the distance from a vertex in comparison to its distance from the opposite side's midpoint.*



### Compare and Connect

Students have an opportunity to compare the distance from the vertex to the centroid to the distance from the centroid to the midpoint. After constructing the centroid, prompt students to mark each distance using patty paper. Ask students what relationship they notice and prompt them to test their conjectures with each median in the triangle.



Consider creating and displaying an anchor chart with the definition of median and centroid of a triangle, or adding the terms to a classroom Word Wall.



## 11.7.3 Guided Instruction: Centroid of a Triangle

**Component Overview:** Students discover that the length from the vertex to the centroid is twice the length from the centroid to the midpoint.



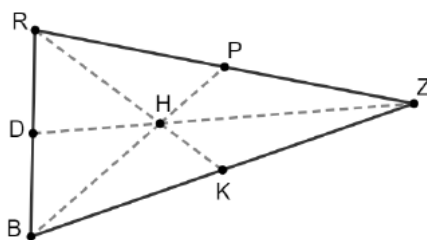
### 11.7.3 Guided Instruction: Centroid of a Triangle



The length from the vertex to the centroid is twice the length from the centroid to the midpoint of the opposite side, which yields a ratio of 2:1.

1.  $\triangle RZB$  is shown, where point  $H$  is the centroid of the triangle.

$DH = 11.3$ ,  $RK = 20.55$ , and  $BH = 15.2$ .



- a. Complete the table.

|             |             |
|-------------|-------------|
| $HZ = 22.6$ | $DZ = 33.9$ |
| $RH = 13.7$ | $HK = 6.85$ |
| $HP = 7.6$  | $BP = 22.8$ |

- b. If  $BK = 6x + 3$  and  $KZ = 10x - 7$ , what is the length of  $\overline{BZ}$ ?

36

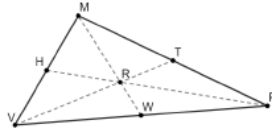


This component will consolidate the learning from 11.7.2 Exploration. Consider prompting students to use colored pencils to mark their diagrams while you read the text in the blue box aloud. For example, the segments from the vertex to the centroid could be highlighted one color, and the segments from the centroid could be highlighted in another color. Students could also label the segment from the centroid to the midpoint with a variable and the segment from the centroid to the vertex with two times the variable. Model how to find a few of the measurements in the table, and then release students to work on the remaining parts independently.



## 11.7.3 Guided Instruction: Centroid of a Triangle (continued)

2.  $\triangle MPV$  is shown, where point  $R$  is the centroid of the triangle.  $MR = 5x - 9$ ,  $RP = 3w + 7$ ,  $RV = 12y - 9$ ,  $RW = 1.75x$ ,  $RT = 4y + 2$ , and  $HR = w + 9$ .



- a. Find the value of each variable.

|         |            |          |
|---------|------------|----------|
| $x = 6$ | $y = 3.25$ | $w = 11$ |
|---------|------------|----------|

- b. Complete the table.

|             |           |             |
|-------------|-----------|-------------|
| $HR = 20$   | $RP = 40$ | $HP = 60$   |
| $RT = 15$   | $VR = 30$ | $VT = 45$   |
| $RW = 10.5$ | $MR = 21$ | $MW = 31.5$ |



### Think-Pair-Share

Prompt students to initially solve the problems individually. Next, ask students to discuss how they obtained their answers with a partner. Finally, call on a variety of students to share their strategies with the whole class. Ask students if they agree or disagree with their peers' strategies and why.

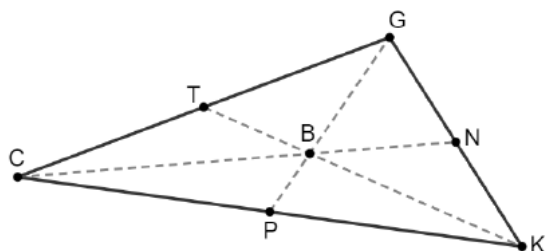
## 11.7.4 Your Turn!

**Component Overview:** Students practice solving problems involving medians and centroids. Students also analyze errors in a sample of student work.



### 11.7.4 Your Turn!

1.  $\triangle CGK$  is shown, where point  $B$  is the centroid of the triangle.  $TB = \frac{1}{2}x - 4$ ,  $PB = 4.4$ ,  $CB = y + 6$ ,  $BK = 2x - 31$ , and  $BN = \frac{1}{3}y + 5$ .



- a. What is the value of  $x$ ?

$$x = 23$$

- b. What is the value of  $y$ ?

$$y = 12$$

- c. Complete the table.

|            |             |
|------------|-------------|
| $TB = 7.5$ | $BK = 15$   |
| $CB = 18$  | $CN = 27$   |
| $BG = 8.8$ | $PG = 13.2$ |

- d. If  $CT = 4k - 10$  and  $TG = 2k + 1$ , what is the length of  $\overline{CG}$ ?

$$24$$



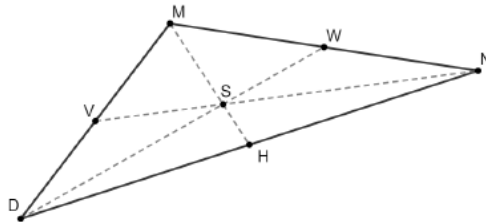
Provide a visual entry point by prompting students to color code and label their diagrams.



- How did you find the value of  $x$  and the value of  $y$ ?
- How can you check your work?

## 11.7.4 Your Turn! (continued)

2.  $\triangle DMN$  is shown, where point  $S$  is the centroid.



- a. Leonardo is finding the value of  $x$  using the information  $VS = \frac{1}{2}x + 1$  and  $SN = 2x - 6$ . His work is shown.

### Leonardo's Work

$$\begin{aligned} 2(2x - 6) &= \frac{1}{2}x + 1 \\ 4x - 12 &= \frac{1}{2}x + 1 \\ \frac{7}{2}x &= 13 \\ x &= \frac{26}{7} \end{aligned}$$

- b. Write a note to Leonardo explaining his mistake.  
*Leonardo,  $\overline{VN}$  is in a 1:2 ratio with  $\overline{VS}$  being the shorter segment. To set up your equation,  $VS$  should be multiplied by 2 and set equal to  $SN$ .*
- c. What is the value of  $x$ ?  
 $x = 8$



Challenge students who finish early to research real-world applications of finding the centroid of a triangle. For example, determining a central location for facilities (i.e. a restroom, restaurant, pool), or finding the center of gravity of a plane or bus to know where to put the fuel and luggage.



Students have the opportunity to demonstrate their mastery of the learning target by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors themselves in the future.

## 11.7.5 Wrap-Up: Cool Down

**Component Overview:** Students write a note to a hypothetical absent student that summarizes the lesson learning targets.



### 11.7.5 Wrap-Up: Cool Down

1. Write a short note summarizing today's lesson to a friend who missed class.

*Answers will vary.*

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*Providing students with sentence stems can help them compose their summary. Consider asking students to create a question to include in their summary.*